Superscripts:

$$2x^3$$

$$2x^{34}$$

$$2x^{3x+4}$$

$$2x^{3x^4+5}$$

Subscripts:

$$x_1$$

$$x_{12}$$

$$x_{1_2}$$

$$x_{1_{2_3}}$$

$$a_0, a_1, a_2, \ldots, a_{100}$$

Greek Letters:

$$\pi$$

$$\Pi$$

$$\alpha$$

$$A = \pi r^2$$

Trig Functions:

$$y = \sin x$$

$$y = \cos x$$

$$y = \csc \theta$$

$$y = \sin^{-1} x$$

$$y = \arcsin x$$

Log Functions:

$$y = \log x$$

$$y = \log_5 x$$

$$y = \ln x$$

Roots:

$$\sqrt{2}$$

$$\sqrt[3]{2}$$

$$\sqrt{x^2 + y^2}$$

$$\sqrt{1 + \sqrt{x}}$$

Fractions:

$$\frac{2}{3}$$

About $\frac{2}{3}$ of the glass is full. About $\frac{1}{3}$ of the glass is empty.
This is display style 1/4: About $\displaystyle\frac{1}{4}$ of the apples are eaten.

$$\frac{\sqrt{x+1}}{\sqrt{x+2}}$$

$$\frac{1}{1 + \frac{1}{x}}$$